

# ALEJANDRO ORTIZ PERDOMO

## AI Engineer | Machine Learning Engineer

📍 Bogotá, Colombia

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| [github.com/alejortizp](https://github.com/alejortizp)

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## PROFESSIONAL SUMMARY

Machine Learning Engineer with end-to-end experience from experimentation to production. Specialized in predictive models (**churn, propensity, NBO**) with quantifiable impact: **Lift 3.0, +15% conversion, +60% contactability**. Proficient in **Scikit-learn, LightGBM, PyTorch** and **Databricks + MLflow** for MLOps. Extensive **deep learning** experience with Transformers and NLP. Combines strong mathematical foundation (Statistics) with modern software engineering to build reliable and scalable ML pipelines.

## EDUCATION

### Master in Software Engineering

[Universidad de los Andes](#)  
2025 – In Progress

### B.S. in Statistics

[Universidad del Valle](#)  
2011 – 2016

### Data Science Diploma (DS4A)

Correlation One, 2022

### Deep Learning Diploma

Univ. Autónoma Occidente, 2023

## LANGUAGES



### Spanish

Native



### English

Professional Proficiency  
(B2)

## KEY SKILLS

Effective Communication

Problem Solving

Analytical Thinking

Team Collaboration

Adaptability

Continuous Learning

Time Management

Technical Leadership

# WORK EXPERIENCE

## AI Engineer — AI Practice

Periferia IT Group

Jun 2025 – Present

- **AI Agent Platform for the Software Development Lifecycle:** Contribute to the technical design, conceptualization and documentation of a proprietary *Single-Responsibility Plug-and-play Agents* (SRPA) ecosystem that automates the end-to-end SDLC: repository analysis and ingestion for developer **onboarding** (ramp-up cut by ~70%), automated **user-story** generation with acceptance criteria, contextual IDE assistance, multi-language **unit testing** with ≥85% coverage, automated QA, dynamic documentation and technical assistants on top of LLMs (**DeepSeek R-1** on **Azure AI Foundry**). Multi-cloud platform (**Azure / AWS / GCP**) used both to accelerate internal delivery and as a commercial offering for clients, delivering **60–70%** savings across the development cycle.
- **CV-Job Matching System:** Pipeline with **Azure Document Intelligence + AI Search** processing 500+ CVs/day with **95%** extraction accuracy and vector semantic search reducing screening time by **70%**. Running in production at **10k+ searches/month** with <200ms latency.
- **Pre-sales & Solution Architecture:** Build technical estimates, team sizing and reference architectures for **GenAI, Agentic AI and RAG** solutions for enterprise clients (banking, insurance, retail, energy and telco). Lead technical discovery with stakeholders and position PerXIA Suite agents in commercial technical proposals.

## Machine Learning Engineer

Global Hitss

Sep 2023 – Jun 2025

- **GenAI & RAG:** Designed and deployed conversational agent with RAG that reduced support tickets by **40%**, processing **10k+ queries/month** with 85% resolution without escalation. Stack: LangChain, Azure OpenAI, FAISS.
- **MLOps:** Implemented CI/CD pipeline in **Databricks + MLflow** reducing deployment time from 2 weeks to **2 days**. Production models with automatic drift monitoring and retraining.
- **Reinforcement Learning:** Productized *Next Best Offer* (NBO) system with A/B testing that increased conversion by **15%** and revenue by **\$10k per person**.
- **Churn Model:** Developed predictive model with **Lift of 3.0** and 82% recall in top decile, integrated to CRM for proactive retention. Impacted **50k+ customers**.

# CERTIFICATIONS

## AI-102: Microsoft Certified Azure AI Engineer Associate

Microsoft, Feb 2026

## NLP Specialization (RAG/Transformers)

Coursera, Jun 2025

## PyTorch Ultimate

Coursera, Mar 2025

## MLOps and Databricks

Netec, Dec 2024

## AWS Cloud Practitioner

Udemy, Feb 2022

# TECHNICAL SKILLS

(years exp.)

## GENAI & LLMS

LangChain 2+ (Agents, RAG)

Azure OpenAI 2+ (GPT-4, Embeddings)

Prompt Engineering 2+

Vector DBs 2+ (FAISS, Pinecone)

## AZURE AI & CLOUD

Azure AI Search 1+

Document Intelligence 1+

Azure AI Foundry 1+

Databricks + MLflow 3+

## GOOGLE CLOUD PLATFORM (GCP)

Vertex AI 1+ (Training, Endpoints, Pipelines)

Cloud Run 1+

Compute Engine 1+ (MIG zonales)

Cloud Storage 1+ | Cloud SQL 1+

## ML/DL FRAMEWORKS

PyTorch 2+ | TensorFlow 3+

Scikit-learn 4+ | XGBoost 4+ | LightGBM 4+

Hugging Face Transformers 2+

## MLOPS & ENGINEERING

Python 7+ | SQL 7+ | PySpark 3+

Docker 2+ | CI/CD 2+

FastAPI 3+ | Streamlit 3+

Git 4+ | GitHub Actions 2+

## Machine Learning Consultant

Aug – Sep 2023

### Grupo CINTE

- **ML Observability:** Designed comprehensive monitoring methodology for *data/model drift* detection with automated retraining alerts, adopted as the data science team's standard.
- **Optimization:** Audited and optimized LightGBM and Random Forest algorithms, improving computational efficiency and reducing inference times.

## Data Scientist

Mar 2023 – Jul 2023

### NAE

- **Contactability Model:** Developed **LightGBM** model that increased effective contact rate by **60%** (from 25% to 40%), optimizing time windows and channels. Impacted campaigns with **100k+ contacts/month**.

## Data Scientist II

May 2022 – Jan 2023

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- **Payment Propensity Model:** Developed predictive model for payment probability and portfolio agreement, maximizing collection through **automated management priorities** based on propensity scores.
- **Recommendation System:** Implemented optimal contact time-slot recommendation engine per account holder, increasing debt collection management effectiveness.
- **Customer Segmentation:** Designed clustering-based marketing strategy for account holders, enabling personalized communication per segment to optimize collection rates.
- **Monitoring:** Built **Power BI** dashboards for tracking collection KPIs and production model performance metrics.

## OPEN SOURCE PROJECTS

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### Email Agent with LangChain

— [github.com/alejortizp/email-agent](https://github.com/alejortizp/email-agent)

Intelligent agent that queries emails via IMAP, classifies with LLM, and generates contextual automatic responses. Reduces response time by **80%**.

**Stack:** LangChain, Azure OpenAI, IMAP, Python, Prompt Engineering

### NL-to-SQL/Pandas Agent — [github.com/alejortizp/nl-to-code](https://github.com/alejortizp/nl-to-code)

Agent that translates natural language to SQL queries and Pandas operations. Includes code validation, secure sandbox execution, and automatic results visualization.

**Stack:** LangChain, SQL, Pandas, Code Interpreter, ReAct Pattern

### RAG Restaurant Ordering System

— [github.com/alejortizp/rag-ordering](https://github.com/alejortizp/rag-ordering)

Conversational ordering system with RAG over restaurant menu. Agent processes product selection in natural language, generates invoices, and calculates totals. Includes inventory management.

**Stack:** LangChain, FAISS, Azure OpenAI, FastAPI, Streamlit

